

# AUTOMATIC SINGING SKILL EVALUATION SYSTEM

Deep Learning-based assessment using CNN-LSTM and engineered features for accurate singing skill evaluation

**92**

Audio Samples

**1-10**

Score Range

**3**

Skill Classes

**4**

Models



Explore the Models



Compare Different  
Models



Check Unknown Audio





# Explore Models



## CNN-LSTM Regression (Original)

Original LSTM version

 Overfitted



## CNN Classification (Simplified)

3-class classifier

 Imbalanced



## Engineered Features + DNN

Hand-crafted musical features

 Best Model



## Engineered Features + K-Fold CV

5-fold cross validation

 Cross-Validated



# CNN-LSTM Regression (Original)

## Predictions

### Model Performance

Test Accuracy: 14.3%  
Overall Accuracy (92 samples): 20.6%

Training (64)

Validation (14)

Test (14)

| Filename         | Actual | Predicted | Status    |
|------------------|--------|-----------|-----------|
| BWxEEnoJaWYk.wav | 5      | 4.59      | ✓ Correct |
| om3YZ-i2234.wav  | 3      | 4.50      | ✗ Wrong   |
| FtvkCWgjXoY.wav  | 3      | 7.63      | ✗ Wrong   |
| Lo9KR3PIVhM.wav  | 5      | 4.65      | ✓ Correct |
| o8eOnfB i5Pc.wav | 7      | 5.87      | ✗ Wrong   |



# CNN Classification (Simplified)

## Predictions

### Model Performance

Test Accuracy: 7.1%  
Overall Accuracy (92 samples): 20.6%

Training (64)

Validation (14)

Test (14)

| Filename        | Actual | Predicted | Status  |
|-----------------|--------|-----------|---------|
| BWxEnoJaWYk.wav | 1      | 0.00      | X Wrong |
| om3YZ-i2234.wav | 1      | 0.00      | X Wrong |
| FtvkCWgjXoY.wav | 1      | 0.00      | X Wrong |
| Lo9KR3PIVhM.wav | 1      | 0.00      | X Wrong |
| q8oQpfRJ5Rc.wav | 2      | 0.00      | X Wrong |



## Engineered Features + DNN

### Predictions

#### Model Performance

Test Accuracy: 50.0%

Overall Accuracy (92 samples): 62.0%

Training (64)

Validation (14)

Test (14)

| Filename        | Actual | Predicted | Status    |
|-----------------|--------|-----------|-----------|
| BWxEnoJaWYk.wav | 1      | 1.00      | ✓ Correct |
| om3YZ-i2234.wav | 0      | 0.00      | ✓ Correct |
| FtvkCWgjXoY.wav | 0      | 0.00      | ✓ Correct |
| Lo9KR3PIVhM.wav | 1      | 1.00      | ✓ Correct |
| q8oQpfRJ5Rc.wav | 1      | 1.00      | ✓ Correct |



## Engineered Features + K-Fold CV

### Predictions

#### K-Fold Results (5 folds)

Fold Accuracies: 42.1%, 47.4%, 38.9%, 44.4%, 38.9%

Mean Accuracy: 42.3%  $\pm$  3.3%

## Model Comparison

Detailed comparison of all four approaches with their architectures and performance

### Dataset Overview

**92**  
Total Samples

**3**  
Skill Classes

**1-10**  
Score Range



 Malayalam & Hindi Melody Songs

### Performance Summary

|                                   |       |
|-----------------------------------|-------|
| Best Accuracy<br>Engineered + DNN | 62.0% |
| Lowest Accuracy<br>CNN-LSTM       | 14.3% |



## CNN-LSTM Regression

Original Architecture

14.3%

Test Acc

20.6%

Overall

### Features

Mel Spectrograms (128×1292)

### Architecture

1. Conv Block 1
- ↓
2. Conv Block 2
- ↓
3. Conv Block 3
- ↓
4. Reshape + Permute
- ↓
5. LSTM Layer
- ↓
6. Dense Layers



## CNN Classification

Simplified CNN

7.1%

Test Acc

20.6%

Overall

### Features

Mel Spectrograms (128×1292)

### Architecture

1. Conv Block 1
- ↓
2. Conv Block 2
- ↓
3. Conv Block 3
- ↓
4. Global Average Pooling
- ↓
5. Dense + Softmax



**Observation:** The model mostly predict class 0. May this is because, it finds





## Engineered Features + DNN

🏆 BEST PERFORMING MODEL

50.0%

Test Accuracy

62.0%

Overall Accuracy

### 17 Engineered Features

#### Spectral Features

- MFCC 1-13 (13 features)
- Spectral Centroid

#### Temporal Features

- Pitch Variance
- Zero Crossing Rate
- Harmonic Ratio

### DNN Architecture

Input Layer (17)



Dense 32 (ReLU) + Dropout 0.3



Dense 16 (ReLU) + Dropout 0.3



Output 3 (Softmax)



**Why it works:** Hand-crafted musical features (17-dim) capture relevant singing characteristics. Simple DNN avoids overfitting with only 1,155 parameters.



## K-Fold Cross Validation

5-Fold on Engineered Features

### Fold Results

|         |                        |       |
|---------|------------------------|-------|
| Fold 1: | <div><div></div></div> | 42.1% |
| Fold 2: | <div><div></div></div> | 36.8% |
| Fold 3: | <div><div></div></div> | 61.1% |
| Fold 4: | <div><div></div></div> | 33.3% |
| Fold 5: | <div><div></div></div> | 55.6% |

### Summary Statistics

45.8%

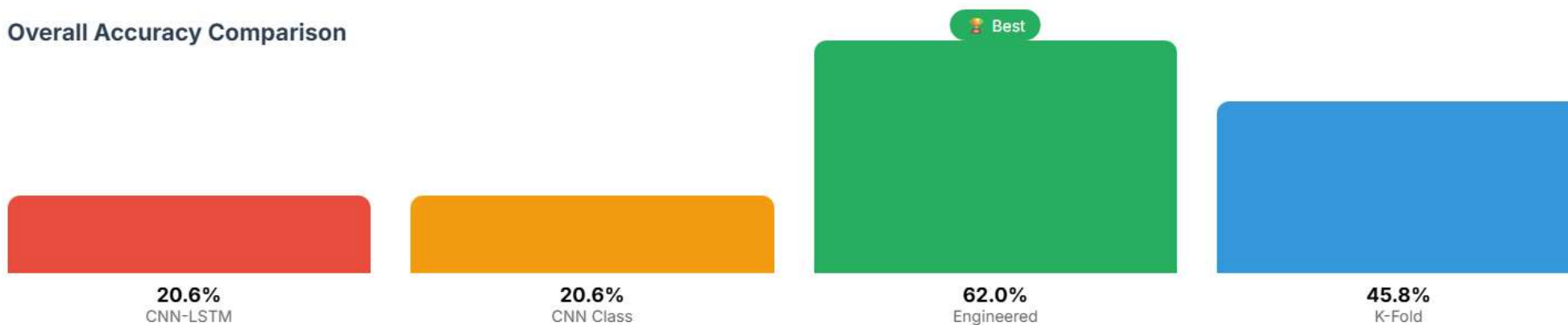
Mean Accuracy

10.8%

Std Deviation

**Most reliable estimate:** Cross-validation provides realistic performance expectation ( $\pm 10.8\%$  variance)

### Overall Accuracy Comparison



## Key Findings & Recommendations



### Best Performer

 WINNER

#### Engineered Features + DNN

17 Features

1,155 params

Test Accuracy

**50.0%**

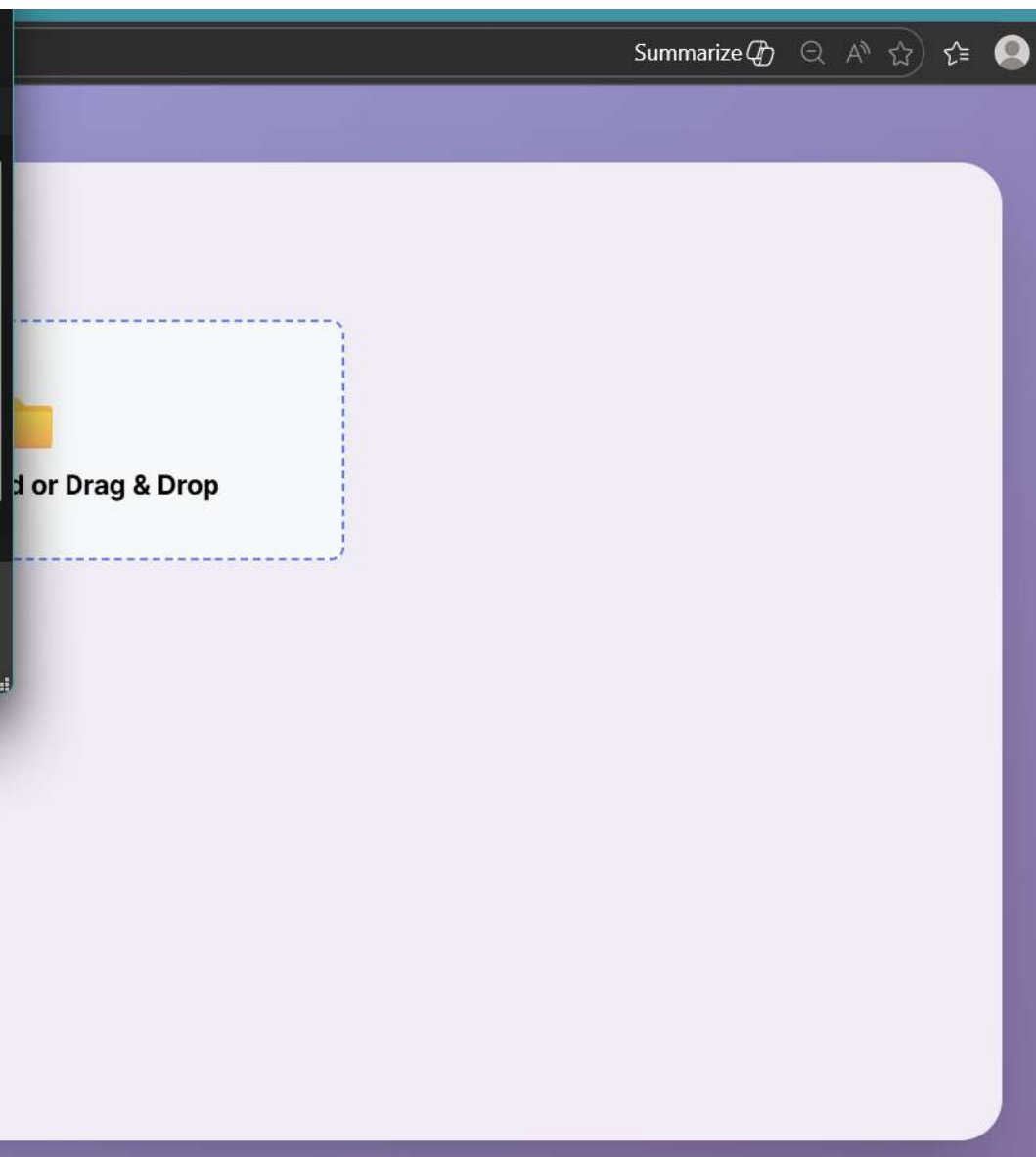
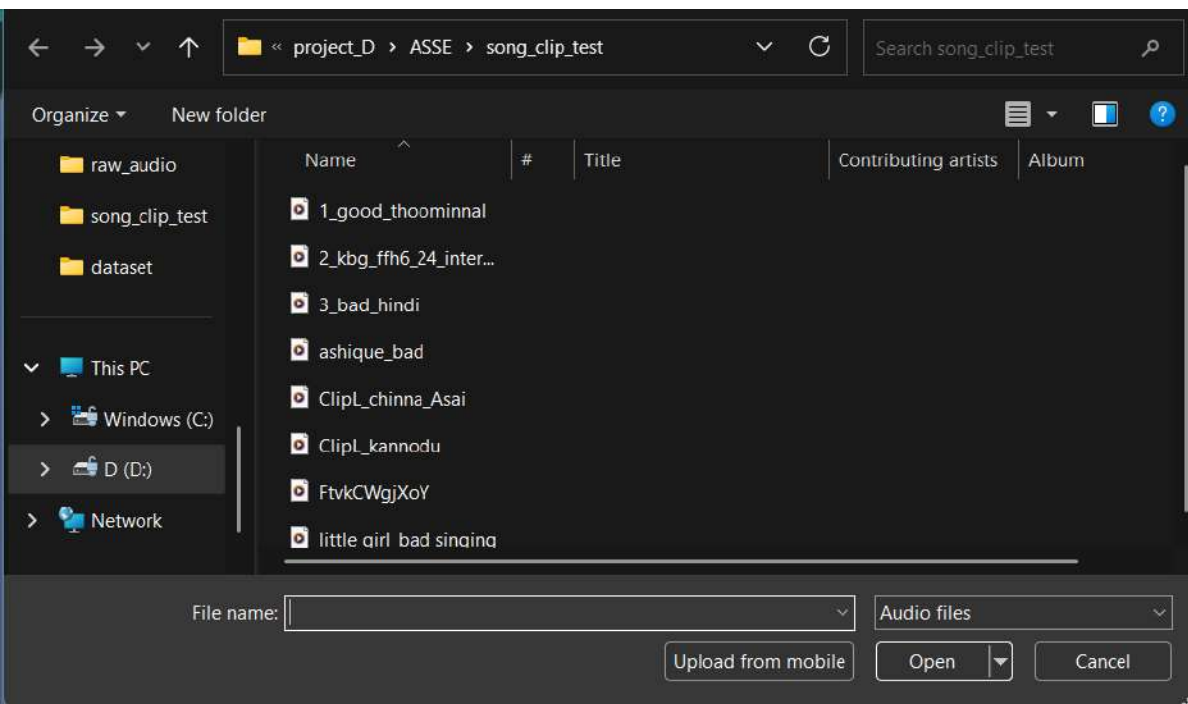
Overall Accuracy

**62.0%**

✓ **Why it works:** Hand-crafted musical features capture relevant singing characteristics while simple DNN avoids overfitting.

### Conclusion

- CNN-LSTM overfits (41,937 params) with 92 samples
- CNN classification struggles to distinguish intermediate and good singing.
- Engineered features + DNN achieves 62.0% accuracy
- K-Fold CV gives realistic estimate:  $45.8\% \pm 10.8\%$





# Test Unknown Audio

## Uploaded Audio



## ANALYSIS RESULT

Skill Level  
**Good (8-10)**

Score  
**9/10**

Confidence  
**48.9%**

## Probability Distribution

